EXPERIMENT – 10

Collecting Email Evidence in Victim PC

Aim

To collect e-mail evidence from a victim's PC by acquiring the physical memory and extracting the e-mail artifacts.

Procedure

- Memory acquisition: The first step in collecting e-mail evidence from the victim's PC is to capture the victim's RAM using the DumpIt tool.
- DumpIt is a utility that generates a physical memory dump of Windows machines; it works on both x86 (32-bit) and x64 (64-bit) systems and is perfect to deploy on a USB key for quick incident response.
- Run dumpit.exe. The destination path of the raw memory dump is displayed and a confirmation question is prompted.
- Type 'Y' and press Enter to proceed; the raw memory dump (.raw / memory dump) is generated and saved in the same directory.
- Feature extraction: Download and install the Bulk Extractor Viewer from GitHub.
- Open Bulk Extractor Viewer and click to generate a report; select the DumpIt memory image as the input and choose an output folder for the report, then click 'Start'.
- Bulk Extractor scans the memory image and writes its output into several feature files such as email.txt, email_histogram.txt, url.txt, telephone.txt, etc.
- Open email.txt to inspect the e-mail addresses saved in the victim system, and open email_histogram.txt to view how often each address occurs.

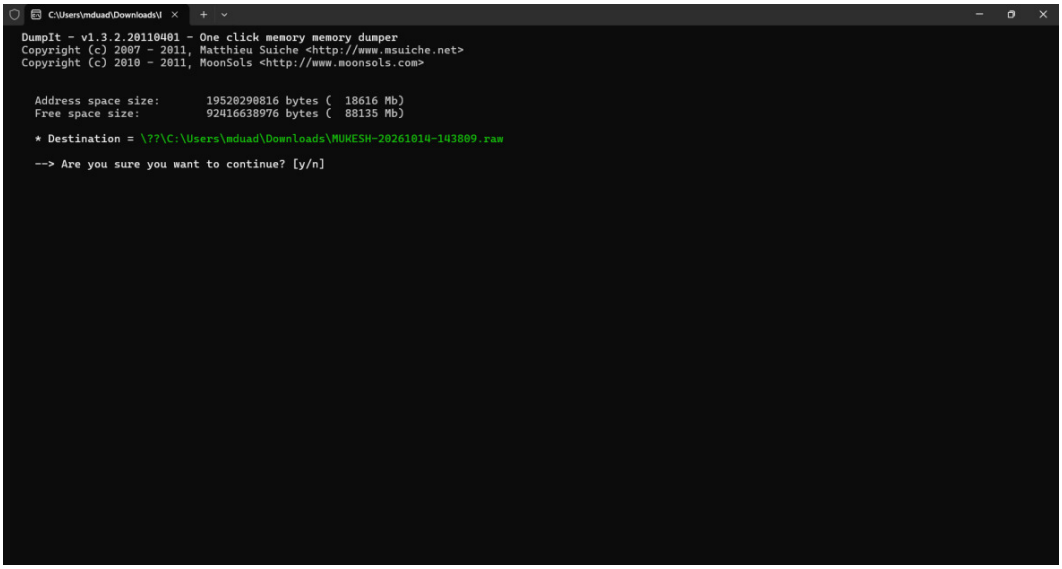


Fig. 10.1 DumpIt – acquiring a raw physical memory dump
Timestamp: 13-10-2026, 15:12:41

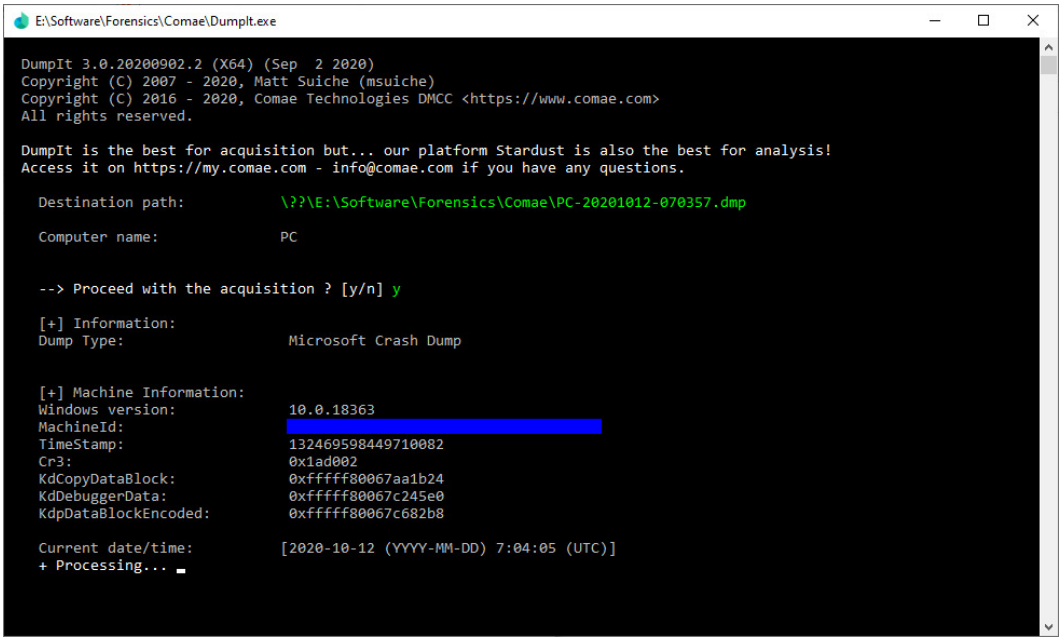


Fig. 10.2 DumpIt – memory acquisition in progress
Timestamp: 13-10-2026, 15:18:07

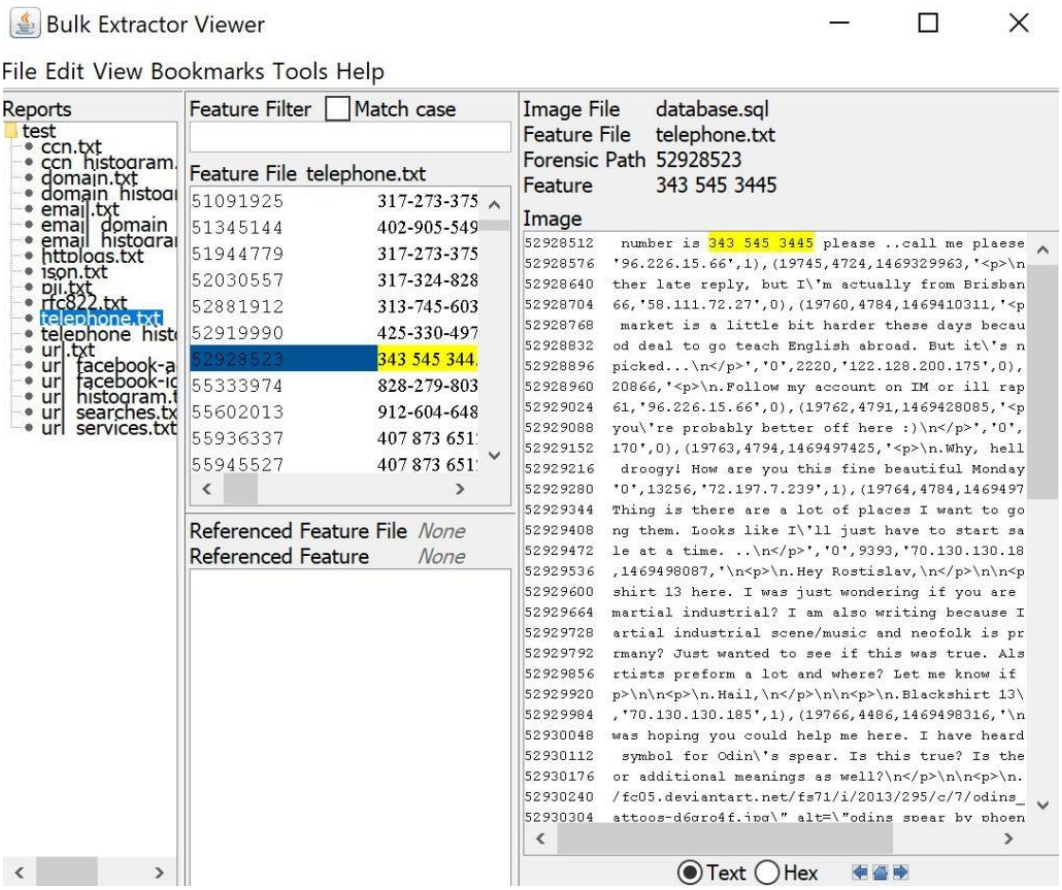


Fig. 10.3 Bulk Extractor Viewer – extracted feature files from the memory image
Timestamp: 13-10-2026, 15:36:28

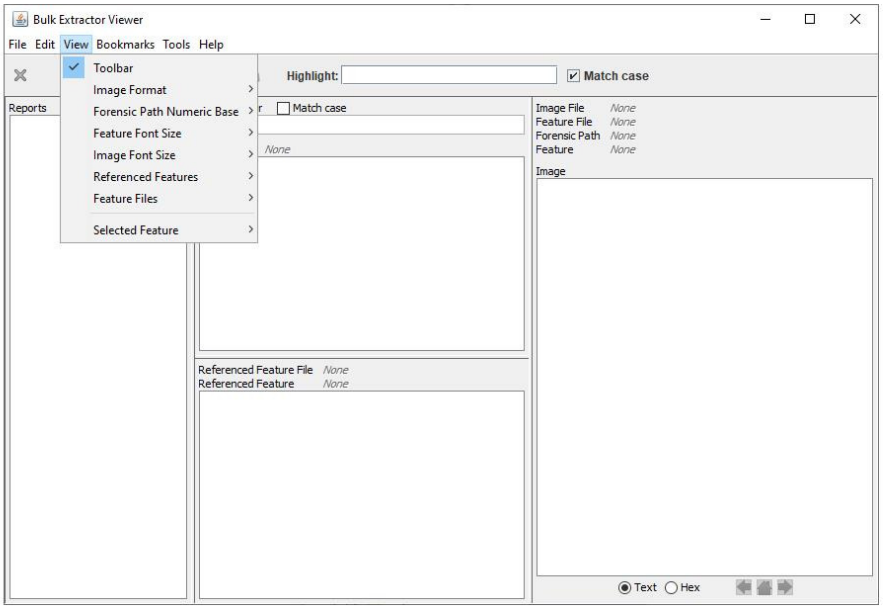


Fig. 10.4 Bulk Extractor Viewer – viewing the extracted features
Timestamp: 13-10-2026, 15:41:55

Result

Thus, the e-mail evidence from the victim's PC was collected successfully. The physical memory of the victim system was acquired using DumpIt, the memory image was processed with Bulk Extractor, and the e-mail addresses and their histogram were extracted and examined.